

INHA UNIVERSITY TASHKENT DEPARTMENT OF CSE & ICE

FALL SEMESTER 2024

SOC 3010 - OPERATING SYSTEMS

TERM PROJECT GUIDELINES FOR TERM PROJECT TITLE SELECTION

TERM PROJECT INVOLVES THE FOLLOWING MAIN TASKS :

- **SELECTION OF A PARTICULAR IPC RELATED PROBLEM**
 - **PROJECT DESIGN & IMPLEMENTATION/SIMULATION**
 - **PREPARATION OF THE PROJECT REPORT**
 - **PRESENTATION OF THE PROJECT WORK**
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- **EACH PROJECT TEAM CAN CONSIST OF 4 TO 8 MEMBERS WITH A MAXIMUM OF 12 PROJECT TEAMS IN A SECTION.**
 - **PROJECT TEAM CAN HAVE MEMBERS DRAWN FROM OTHER SECTIONS ALSO.**
 - **PROJECT TEAMS ARE GIVEN TIME TILL 3RD WEEK OF THE SEMESTER TO SELECT AN IPC (INTER-PROCESS COMMUNICATION) RELATED REAL-TIME PROBLEM PERTAINING TO SOCIETY /SOCIAL NETWORKING.**
 - **TEAM LEADER OF EVERY TERM PROJECT TEAM MUST SELECT A PROJECT TITLE AND SEND TO MY E-MAIL DIRECTLY THE SELECTED PROJECT TITLE ALONG WITH THE TEAM DETAILS IN THE ATTACHED FILE TEMPLATE PROVIDED AT THE ECLASS PORTAL.**

- **DEADLINE FOR RECIEVING THE TERM PROJECT TITLE ALONG WITH THE TERM PROJECT TEAM DETAILS IS FRIDAY, 27th SEPTEMBER 2024.**
- **IF POSSIBLE, PROVIDE THREE PREFERENCES IN THE ORDER 1, 2, 3. IF PREFERENCE 1 IS ALREADY ALLOTTED THEN ATLEAST NEXT PREFERENCE CAN BE ALLOTTED WITHOUT WAITING FOR NEXT ROUND OF ALLOTTMENT**
- **PROJECT TITLE WILL BE ALLOTTED ON A FIST-COME FIRST-SERVE BASIS**
- **NO TWO TEAMS CAN HAVE THE SAME TERM PROJECT TITLE CONSIDERING ALL SECTIONS**

SOME PROJECT TITLES PERTAINING TO DIFFERENT SECTORS ARE GIVEN BELOW: (BUT THIS IS NOT AN EXHAUSTIVE LIST)

- 1. BANKING SYSTEM WITH ALL BANKING FACILITIES LIKE – DEPOSIT, WITHDRAWAL, FOREIGN EXCHANGE TO ANY CURRENCY, AVAILABILITY OF LOANS FOR PURCHASING VEHICLES, APARTMENTS, HOUSES, SETTING UP BUSINESS, EDUCATION LOAN, MANAGEMENT OF ATMs AND ALL OTHER FEATURES.**

2. **AIRLINE FLIGHT RESERVATION SYSTEM**
(ONLINE BOOKING OF TICKETS IN DIFFERENT FLIGHTS FOR DIFFERENT DESTINATIONS ALL OVER THE WORLD, CANCELLATION OF TICKETS, CLEAR DISPLAY OF CANCELLATION AMOUNT, REFUND OF AMOUNT AFTER CANCELLATION, SHOWING AVAILABILITY OF ALL FLIGHTS, SHOWING FLIGHTS TIMINGS FOR ALL 7 DAYS OF A WEEK, SEATS AVAILABILITY, SEAT SELECTION FOR TRAVELLERS BY GIVING THE COMPLETE LAYOUT OF THE SEATING ARRANGEMENT INSIDE THE FLIGHTS, FOOD AVAILABILITY/NON-AVAILABILITY INSIDE THE FLIGHTS, CHANGE OF TRAVEL DATES AND AMOUNT CHARGED)
3. **TAXI/CAB SHARING**
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)
4. **UZBEKISTAN UNIVERSITY EDUCATION PORTAL**
(PROVIDING ALL INFORMATION ABOUT UNDER-GRADUATE, POST GRADUATE AND DOCTORAL PROGRAMMES OFFERED, FACILITIES AVAILABLE, LOCATION & MAP, FEE STRUCTURE IN ALL THE UNIVERSITIES IN UZBEKISTAN)
5. **ONLINE EXAM MANAGEMENT SYSTEM**
(WITH TOTAL SECURITY OF IDENTIFYING THE STUDENTS DURING EXAM, MONITORING THE STUDENTS ACTIVITIES DURING THE EXAM, SELECTION OF DIFFERENT QUESTIONS FOR EACH STUDENT, DEVELOPMENT OF A LARGE QUESTION BANK CONTAINING HUNDREDS OF QUESTIONS IN EACH SUBJECT CONSIDERING ALL COURSES TAUGHT AT THE UNIVERSITY)

6. **LIBRARY MANAGEMENT SYSTEM**
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)
7. **E-CONTENT MANAGEMENT SYSTEM**
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)
8. **PLAGIARISM CHECKER & FILE MANAGEMENT SYSTEM**
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)
9. **UZBEKISTAN HOTEL RESERVATION & MANAGEMENT PORTAL**
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)
10. **UZBEKISTAN RESTAURANT MANAGEMENT**
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)
11. **HEALTHCARE CONSULTING SYSTEM (DOCTORS WITH DIFFERENT SPECIALIZATIONS FOR CONSULTATION, HOSPITALS WITH ALL FACILITIES FOR TREATING DIFFERENT DISEASES IN UZBEKISTAN & ABROAD – ONE STOP PORTAL FOR ALL CONSULTATIONS AND TREATMENTS)**
12. **ELECTRONIC HEALTH RECORD MANAGEMENT SYSTEM WITH BUILTIN SECURITY**
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)

- 13. PHARMACY- MEDICAL STORE MANAGEMENT
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 14. BLOOD BANK SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 15. ONLINE SHOPPING AND DELIVERY SYSTEM
(SIMILAR TO AMAZON)
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 16. ONLINE CAR SHOPPING
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 17. UZBEKISTAN TOURISM PORTAL
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 18. WORLD TOURISM PORTAL
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 19. HIGHER EDUCATION ABROAD PORTAL
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 20. JOB SEARCH/RECRUITMENT PORTAL
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 21. IUT RESOURCE MANAGEMENT SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**

- 22. ATTENDANCE MONITORING SYSTEM WITH FINGERPRINTS VERIFICATION
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 23. FACE RECOGNITION BASED ATTENDANCE CHECKING SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 24. AIRCRAFT COMMUNICATION AND MONITORING SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 25. TICKET BOOKING MANAGEMENT SYSTEM FOR CONCERT CEREMONIES
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 26. ALL STORE STOCK MANAGEMENT (INVENTORY CONTROL)
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 27. MULTIPLAYER GAMING APPLICATIONS
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 28. CITY TRAFFIC MONITORING AND CONTROL SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**

- 29. POLICE TRAFFIC VIOLATION REPORTING & CONTROL SYSTEM.
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 30. WORLD VISA PORTAL
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 31. THE MARRIAGE FUNCTION HALL BOOKING & FOOD/MUSIC ARRANGEMENT SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 32. ANY VEHICLE (CAR, BUS, HEAVY VEHICLES FOR PARTIES, FUNCTIONS, FAMILY PICNICS, LONG DISTANCE TRAVEL) BOOKING PORTAL
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 33. TEACHERASSISTED PROGRAM WRITING ENVIRONMENT FOR STUDENTS
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 34. DOCTORS RESERVATION SYSTEM FOR PATIENTS
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 35. BUS RESERVATION & TRACKING SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 36. RAILWAY BOOKING AND TRAIN TRACKING SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**

- 37. WAREHOUSE MANAGEMENT SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 38. COURIER TRACKING, CARGO AND FREIGHT
TRANSPORTATION
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 39. ONLINE CODE TESTING SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 40. INHA ONLINE QUIZ SYSTEM
(WITH TOTAL SECURITY OF IDENTIFYING THE
STUDENTS DURING THE QUIZ, MONITORING THE
STUDENTS ACTIVITIES DURING THE QUIZ,
SELECTION OF DIFFERENT QUIZ QUESTIONS FOR
EACH STUDENT, DEVELOPMENT OF A LARGE QUIZ
QUESTION BANK CONTAINING HUNDREDS OF QUIZ
QUESTIONS IN EACH SUBJECT CONSIDERING ALL
COURSES TAUGHT AT THE UNIVERSITY)**
- 41. UZBEKISTAN LAND/HOUSE/APARTMENT RENTAL &
PURCHASE PORTAL
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**
- 42. HOUSECLEANING, PLUMBING, ELECTRICITY
SERVICE & MAINTENANCE SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED
LATER, IF NEEDED)**

- 43. HUMAN ORGAN TRANSPLANTATION MANAGEMENT SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 44. COVID-19 TRACKING, TESTING, TREATMENT & HOSPITAL MANAGEMENT SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 45. MULTI-CRYPTOCURRENCY TRADING PORTAL (EXCHANGE) ALLOWING TRADING OF ALL CRYPTO COINS USING SECURITY, CONFIDENTIALITY AND AUTHENTICATION AND WALLET CREATION FOR EACH CUSTOMER FOR TRANSACTIONS
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 46. PARKING MANAGEMENT SYSTEM
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 47. ONLINE FOOD DELIVERY SYTEM
(LINKED TO ALL RESTAURANTS IN DIFFERENT DISTRICTS IN TASHKENT, SAMARKAND, BHOKARO AND OTHER REGIONS IN UZBEKISTAN)
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**
- 48. STUDENT ATTENDANCE CHECKING USING PHONE'S NFC (NEAR-FIELD COMMUNICATION)
(DETAILED DESCRIPTION WILL BE PROVIDED LATER, IF NEEDED)**

OR

- **APPLICATIONS PERTAINING TO SOCIAL MEDIA OR ANY OTHER DOMAIN OF INTEREST USEFUL TO THE SOCIETY AND CARRY OUT THE PROJECT WORK IN A TEAM.**
- **EACH PROJECT TEAM IS REQUIRED TO DEVELOP THE SOFTWARE THAT MODEL THE BUSINESS PROCESS ACCORDING TO THE SYSTEM REQUEST. IT MUST BE IMPLEMENTED USING CLIENT – SERVER ARCHITECTURE WITH SERVER ON ONE MACHINE AND CLIENTS IMPLEMENTED ON DIFFERENT MACHINES.**
- **ALL INTER PROCESS COMMUNICATIONS BETWEEN SERVER AND CLIENTS SHOULD BE ORGANIZED USING THE IPC MECHANISM - SOCKETS (SOCKETS PROGRAMMING WITH MULTIPLE CLIENTS ON DIFFERENT MACHINES).**
- **THE INTER PROCESS COMMUNICATION BETWEEN SERVER AND CLIENTS USING SOCKETS SHOULD BE IMPLEMENTED USING C LANGUAGE.**
- **THE FRONTEND GUI MUST BE DEVELOPED USING THE GUI TOOLS AVAILABLE ON THE LINUX SYSTEM. (USING GLADE, GTK+3 OR ANY OTHER TOOLS OF YOUR CHOICE)**
- **BACKEND DATABASE MUST BE DEVELOPED USING ANY ONE OF THE FAMILIAR DATA BASE MANAGEMENT SYSTEM TOOLS AVAILABLE ON THE LINUX SYSTEM.**
- **KNOWLEDGE OF C PROGRAMMING, JAVASCRIPT/ JASON SCRIPT, MySQL AND GUI TOOLS ON LINUX IS HIGHLY DESIRABLE**

➤ **THE TITLE OF THE SELECTED TERM PROJECT ALONG WITH ABSTRACT, BRIEF PROJECT DESCRIPTION, REQUIREMENTS DEFINITION DOCUMENT IN THE FORM OF A PROJECT PROPOSAL (PRELIMINARY DOCUMENT) MUST BE SUBMITTED BY THE END OF 6TH WEEK OF THE SEMESTER (THURSDAY 17TH OCTOBER 2024).**

➤ **PREPARE THE PROJECT PROPOSAL USING THE TEMPLATE PROVIDED AT THE ECLASS PORTAL.**

➤ **EVERY MEMBER OF THE TEAM MUST UPLOAD THE SOFTCOPY OF THE PROJECT PROPOSAL AT THE ECLASS PORTAL**

➤ **LAST DATE FOR SUBMISSION OF THE PROJECT PROPOSAL IS THURSDAY, 17TH OCTOBER 2024**

➤ **PROJECT PROPOSAL PREPARED USING THE TEMPLATE WILL ONLY BE ACCEPTED**

➤ **LATE SUBMISSIONS ARE NOT ENTERTAINED, ADHERE TO THE DEADLINE STRICTLY**

➤ **NO TWO TEAMS (CONSIDERING ALL THE SECTIONS) CAN HAVE THE SAME PROJECT.**

➤ **THE PROBLEM SELECTED MUST BELONG TO DIFFERENT AREAS.**

- **PRESENTATION WILL BE HELD IN THE 16TH WEEK OF THE SEMESTER (23RD TO 28TH DECEMBER 2024 – EXACT SCHEDULE WILL BE ANNOUNCED LATER)**
- **MAXIMUM DURATION OF THE PRESENTATION WILL BE 45 MINUTES FOLLOWED BY 15 MINUTES FOR QUESTION/ANSWER SESSION FOR EACH PROJECT TEAM. ALL MEMBERS OF THE TEAM MUST PRESENT A PART OF THE PROJECT CARRIED-OUT IN ROTATION. THE TIME-DIVISION FOR PRESENTATION AMONG THE MEMBERS MUST BE DONE JUDICIOUSLY SO THAT COMPLETE PROJECT WORK CARRIED OUT IS PRESENTED IN A PERFECT WAY WITHOUT MISSING ANY DETAIL. A LIVE DEMO OF THE PROJECT DEVELOPED IMMEDIATELY AFTER THE PRESENTATION – DURATION OF DEMO IS 30 MINUTES.**
- **THE FINAL PROJECT REPORT MUST INCLUDE THE FOLLOWING SECTIONS : ABSTRACT, INTRODUCTION, PROJECT OVERVIEW, REQUIREMENTS DEFINITION SECTION INCLUDING USE CASE DIAGRAM, ACTIVITY DIAGRAMS, CLASS DIAGRAMS, BEHAVIORAL DIAGRAM, PROJECT DESIGN & IMPLEMENTATION, SIMULATION RESULTS & DISCUSSIONS, CONCLUSION, FUTURE WORK, REFERENCES. FINAL PROJECT REPORT TEMPLATE WILL BE PROVIDE AT THE ECLASS PORTAL. THE PROJECT REPORT MUST BE SUBMITTED ATLEAST ONE DAY BEFORE THE**

**PRESENTATION OR JUST BEFORE THE
PRESENTATION ON THE DAY OF THE
PRESENTATION**

**➤ PROJECT DELIVERABLES: PROJECT REPORT
(ONLY SOFT COPY) WITH SIMULATION RESULTS,
SOURCE CODE DEVELOPED, PRESENTATION PPTS.**
